

Northwestern University
CIERA: Center for Interdisciplinary
Exploration and Research in Astrophysics
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James W. Johnson
Curriculum Vitae
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ACADEMIC POSITIONS

Northwestern University

Evanston, Illinois

2026 – present **Postdoctoral Associate**

Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA)
LSST-DA Data Science Fellowship Program
Supervisor: Prof. Adam A. Miller

Carnegie Institution of Washington – The Observatories

Pasadena, California

2023 – 2026 **Postdoctoral Fellow**, Carnegie Theoretical Astrophysics Center (CTAC)
Supervisor: Dr. Ana Bonaca

EDUCATION

The Ohio State University

Columbus, Ohio

July 2023 **Ph.D. in Astrophysics**, Dissertation Advisor: Prof. David H. Weinberg
*From Dwarfs to Spirals: Chemical Evolution of Galaxies across Stellar Mass
and the Implications for Nucleosynthesis*

November 2019 **M.S. in Astrophysics**

Vanderbilt University

Nashville, Tennessee

May 2017 **B.A.**, Physics (major), Astronomy (minor), *cum laude*
Highest Honors in Astronomy, Thesis Advisor: Prof. Andreas A. Berlind

RESEARCH

39	14	25	1,100+	19
Journal Publications	1st & 2nd Author	Contributing Author	Citations	H-Index

Interests

Galactic chemical evolution – The Milky Way – Dwarf galaxies – The astrophysical origin of the elements – Big bang nucleosynthesis – Near field cosmology – Astronomical software

[NASA ADS Libraries](#) (A full list of my journal publications is included.)

All My Papers <https://ui.adsabs.harvard.edu/public-libraries/rIqfpNKmSdaOMIAhkk2VzQ>
1st & 2nd Author <https://ui.adsabs.harvard.edu/public-libraries/go1WSseGTMeft2SxdESAgw>
Co-Author https://ui.adsabs.harvard.edu/public-libraries/sZkjSf_XRSKSRykqBe6B_w

Seminars & Conference Presentations

Invited Talk **APOGEE's Galactic Odyssey** 2026
Centro de Estudios de Fisica del Cosmos de Aragón (Teruel, Spain)
Invited Talk **Galactic Metal Cycles**, University of Notre Dame (Kylemore Abbey, Ireland) 2026

Invited Seminar	University of California, Davis , Dept. of Physics and Astronomy (Davis, CA)	2025
Invited Seminar	Stockholms Universitet , Dept. of Astronomy (Stockholm, Sweden)	2025
Invited Seminar	Uppsala Universitet , Dept. of Physics & Astronomy (Uppsala, Sweden)	2025
Poster	Small Galaxies, Cosmic Questions - II	2024
	University of Durham (Durham, United Kingdom)	
Invited Talk	DHWFEST: Dark, Hot, Warm, and Fuzzy mattEr in Space and Time	2024
	University of Utah (Salt Lake City, UT)	
Invited Seminar	Lund University , Dept. of Physics (Lund, Sweden)	2024
Contributed Talk	ADONIS: Abundance Gradients in the Local Universe	2024
	Munich Institute for Astro-, Particle, and BioPhysics (MIAPbP) (Munich, Germany)	
Contributed Talk	Surveying the Milky Way: The Universe in Our Own Backyard	2023
	California Institute of Technology (Pasadena, CA)	
Dissertation Talk	241st American Astronomical Society Meeting (Seattle, WA)	2023
Contributed Talk	GALactic Archaeology with Hermes (GALAH) Science Meeting	2021
	(virtual)	
Poster	236th American Astronomical Society Meeting (virtual)	2020
Invited Seminar	University of California, Santa Cruz (Santa Cruz, CA)	2019
	Sloan Digital Sky Survey (SDSS) Collaboration Meetings	
Contributed Talk	Katholieke Universiteit Leuven (Leuven, Belgium)	2026
Contributed Talk	Max Planck Institut für Astronomie (Heidelberg, Germany)	2025
Contributed Talk	New Mexico State University (Las Cruces, NM)	2024
Contributed Talk	Virtual	2021
Contributed Talk	Virtual	2020

Astrophysical Software Development



Versatile Integrator for Chemical Evolution (VICE)

Lead developer and license owner (Spring 2018 – Present)

Documentation: <https://vice-astro.readthedocs.io>

Source Code: <https://github.com/giganano/VICE.git>

Install: <https://pypi.org/project/vice>

Observing Programs

PI: *The First Extragalactic Measure of the Helium Isotopic Ratio – A New Test of Fundamental Physics*

2024B WINERED spectrograph, 18 hours (Clay 6.5-m Telescope, Las Campanas Observatory)

2025A MIKE spectrograph, 6 hours (Clay 6.5-m Telescope, Las Campanas Observatory)

MENTORSHIP

Cal-Bridge Summer Research Program

- 2025 – Present **Christopher Giudice** (undergraduate), San Francisco State University
 Project: The Chemical Equilibration Timescale of the Milky Way Disk
 Now: M.Sc. student at California State University, Long Beach
- 2024 – 2025 **Damien Tessmer** (undergraduate), San Diego State University
 Project: Identifying trends in stellar nucleosynthesis with SDSS-V data

The Ohio State University

- 2021 – Present **Liam O. Dubay** (graduate), Dept. of Astronomy
 Projects: Galactic chemical evolution in the Milky Way
 (arxiv:2404.08509, 2508.00988, 2608.26261)
- 2022 – Present **Daniel A. Boyea** (undergraduate), Dept. of Astronomy
 Project: Investigating the astrophysical origin of carbon
 Now: Ph.D. student at University of Groningen (Groningen, Netherlands)

- 2021 – 2023 **Miquela K. Weller** (graduate), Dept. of Astronomy
Projects: Investigating the astrophysical origin of helium (arxiv:2404.08765)
- 2022 – 2023 **Lindsey Stultz** (undergraduate), Dept. of Physics
Polaris Near-Peer Mentorship Program

HONORS & AWARDS

- 2023 **CTAC Postdoctoral Fellowship**, Carnegie Science
- 2022 **Ann S. Tuttle Paper Prize**, Ohio State, Dept. of Astronomy
Annual award to the top graduate student-led publication of the previous year
Paper: Johnson J.W., et al., 2021, MNRAS, 508, 4484 (arxiv:2103.09838)
- 2022 – 2023 **Presidential Fellowship**, Ohio State, College of Arts & Sciences
Financial support for final-year graduate students
- 2017 – 2018 **University Fellowship**, Ohio State, College of Arts & Sciences
Financial support for first-year graduate students
- 2017 **Larry Ross Cathey Award**, Vanderbilt, Dept. of Physics & Astronomy
Outstanding graduating senior studying astronomy
- Inducted 2015 **Sigma Pi Sigma Physics National Honor Society**, Vanderbilt Chapter
7 of 8 semesters **Dean's List**, Vanderbilt, College of Arts & Sciences

COMMUNITY INVOLVEMENT

Carnegie Science Observatories

- 2024 – Present **Research Mentor**, Cal-Bridge Summer Research Program (2 students)
- 2024 **Advancing Inclusive Mentoring**
12+ hours of instruction and discussion on equitable mentorship practices

Polaris Near-Peer Mentorship Program

Graduate student-leg organization at Ohio State dedicated to fostering a more inclusive environment and improving retention of underrepresented minority groups in the Dept. of Physics and the Dept. of Astronomy. Website: <https://u.osu.edu/polaris>.

- 2022 – 2023 **Leadership Committee** (budget: ~\$60,000/year)
- 2022 **Academic Facilitator**, Undergraduate Residential Summer Access Program
Early-arrival program for first-year undergraduate students
- 2022 – 2023 **Near-Peer Mentor** (1 student)

TEACHING

Undergraduate Python Coding Workshop

- Program Creator**, six sessions, ~25 hours of instruction and exercises
Website: <https://jamesjohnson.space/bootcamp>
Source material: <https://github.com/giganano/PythonBootcamp>
- 2020 – 2023 **Full program (annually)**: Summer undergraduate research students
The Ohio State University, Dept. of Astronomy
- 2022 **Full program**: 1st- & 2nd-year graduate students
The Ohio State University, Dept. of Astronomy
- 2024 **Select sessions**: CASSI & Cal-Bridge undergraduate research students
Carnegie Science Observatories

The Ohio State University, Department of Astronomy: Graduate Teaching Assistant

2018 – 2020	Astronomy 1101: From Planets to Cosmos	5 sections
2019	Astronomy 1142: Black Holes	1 section
2019	Astronomy 1221: Astronomy Data Analysis	1 section
2018	Astronomy 1140: Planets and the Solar System	1 section

MISCELLANEOUS

- 2022 – Present **Manuscript Referee:** ApJ, MNRAS, PASJ, A&A
- 2024 – 2026 **Working Group Co-Chair**, Galactic Genesis, Sloan Digital Sky Survey-V
- 2024 – 2026 **“Morning Tea” co-organizer** (daily arXiv discussion), Carnegie Science
- 2024 – 2025 **External Panelist**, Hubble Space Telescope Proposal Review, Cycles 32 & 33
- 2021 – 2023 **“Galaxy Hour” meeting organizer**, Ohio State, Dept. of Astronomy
- 2017 – 2023 **Diversity Journal Club**, Ohio State, Dept. of Astronomy
- June 2020 **Real Scientists Germany Online Outreach**
 Blog: <https://tinyurl.com/jamesjohnsonrealscientistsDE>
 Twitter: https://twitter.com/realsci_DE
- 2015 – 2017 **Undergraduate Tutor, Proctor, Grader**
 Vanderbilt University, Dept. of Physics & Astronomy
- 2015 **Cosmic Ray Observatory Project**, Instrumentation lab
 University of Nebraska-Lincoln, Dept. of Physics

JOURNAL PUBLICATIONS

First & Second Author (reverse chronological order)

1. *Towards a measurement of the primordial helium isotope ratio*
 Cooke R.J., **Johnson J.W.**, Noterdaeme P, Pettini M., Welsh L., Wong A., Peroux C.
 2026, ApJ, 1003, 225 – 240 arxiv:2605.00122
2. *The galactic chemical evolution of carbon: Implications for stellar nucleosynthesis*
 Boyea D.A., **Johnson J.W.**, Weinberg D.H.
 2025, MNRAS, 550, 1155 – 1174 arxiv:2511.20752
3. *Metals versus Non-Metals: Chemical Evolution of Hydrogen and Helium Isotopes in the Milky Way*
Johnson J.W., Weller M.K., Cooke R.J.
 2025, submitted to AAS Journals, under peer review arxiv:2510.08689
4. *That’s so Retro: The Gaia-Sausage-Enceladus Merger Trajectory as the Origin of the Chemical Abundance Bimodality in the Milky Way Disk*
Johnson J.W., Feuillet D.K., Bonaca A., de Brito Silva D.
 2025, accepted for publication in AAS Journals arxiv:2510.08688
5. *Constraints on Radial Gas Flows in the Milky Way Disk Revealed by Large Stellar Age Catalogs*
Johnson J.W.
 2025, accepted for publication in AAS Journals arxiv:2510.05223
6. *A Galactic Perspective on the (Unremarkable) Relative Refractory Depletion Observed in the Sun*

- Rampalli R., **Johnson J.W.**, Ness M.K., Edwards G.H., Newton E.R., Griffith E.J., Bedell M., Wang K.
2025, ApJ, 1002, 127 – 147 arxiv:2509.03577
7. *Rising from the Ashes II: The Bar-driven Abundance Bimodality in the Milky Way*
Beane A., **Johnson J.W.**, Semenov V., Hernquist L., Chandra V., Conroy C.
2024, ApJ, 985, 221 – 233 arxiv:2410.21580
 8. *The Milky Way Radial Metallicity Gradient as an Equilibrium Phenomenon: Why Old Stars are Metal-Rich*
Johnson J.W., et al.
2024, ApJ, 988, 8 – 35 arxiv:2410.13256
 9. *Dwarf galaxy archaeology from chemical abundances and star formation histories*
Johnson J.W., et al.
2023, MNRAS, 526, 5084 – 5109 arxiv:2210.01816
 10. *Binaries drive high Type Ia supernova rates in dwarf galaxies*
Johnson J.W., Kochanek C.S., Stanek K.Z.
2023, MNRAS, 526, 5911 – 5918 arxiv:2210.01818
 11. *Empirical constraints on the nucleosynthesis of nitrogen*
Johnson J.W., Weinberg D.H., Vincenzo F., Bird J.C., Griffith E.J.
2023, MNRAS, 520, 782 – 803 arxiv:2202.04666
 12. *Stellar migration and chemical enrichment in the Milky Way disc: a hybrid model*
Johnson J.W., et al.
2021, MNRAS, 508, 4484 – 4511 arxiv:2103.09838
 13. *The impact of starbursts on element abundance ratios*
Johnson J.W., Weinberg D.H.
2020, MNRAS, 498, 1364 – 1381 arxiv:1911.02598
 14. *The secondary spin bias of dark matter haloes*
Johnson J.W., Maller A.H., Berlind A.A., Sinha M., Holley-Bockelmann J.K.
2019, MNRAS, 486, 1156 – 1166 arxiv:1812.02206

Contributing Author (reverse chronological order)

1. *A Broken Clock is Right Twice a Day: [Ce/Mg] is not a Universal Chemical Clock*
Dubay L.O., et al., incl. **Johnson J.W.**
2026, submitted to AAS Journals, under peer review arxiv:2608.26261
2. *The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper*
SDSS Collaboration, et al., incl. **Johnson J.W.**
2026, submitted to AAS Journals, under peer review arxiv:2607.26149
3. *Milky Way Mapper Decoded Abundances II: From patterns to paths*
Ness M.K., et al., incl. **Johnson J.W.**
2026, MNRAS, 550, 960 – 976 arxiv:2605.21735

4. *Milky Way Mapper Decoded Abundances I: Shared disk enrichment patterns*
Ness M.K., et al., incl. **Johnson J.W.**
2026, MNRAS, 548, 637 – 656 arxiv:2605.20487
5. *Scylla at APOGEE: The Impact of Starbursts on the Chemical Evolution of the Magellanic Clouds*
Escala I., et al., incl. **Johnson J.W.**
2026, ApJ, 1003, 237 – 263 arxiv:2603.25909
6. *[C/N] Ages for Red Giants and their Implications for Galactic Archaeology*
Roberts J.D., Pinsonneault M.H., Johnson J.A. Dubay L.O., **Johnson J.W.**
2025, ApJ, 1002, 191 – 206 arxiv:2509.25321
7. *Challenges to the Two-Infall Scenario by Large Stellar Age Catalogs*
Dubay L.O., Johnson J.A., **Johnson J.W.**, Roberts J.D.
2025, ApJ, 1000, 244 – 262 arxiv:2508.00988
8. *The Open Cluster Chemical Abundances and Mapping Survey: VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*
Otto J.M., et al., incl. **Johnson J.W.**
2025, AJ, 171, 91 – 117 arxiv:2507.07264
9. *The Nineteenth Data Release of the Sloan Digital Sky Survey*
SDSS Collaboration, et al., incl. **Johnson J.W.**
2025, ApJS, 285, 9 – 47 arxiv:2507.07093
10. *Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy*
Kollmeier J.A., et al., incl. **Johnson J.W.**
2025, AJ, 171, 52 – 97 arxiv:2507.06989
11. *Many Elements Matter: Detailed Abundance Patterns Reveal Star-formation and Enrichment Differences among Milky Way Structural Components*
Griffith E.J., Hogg D.W., Hasselquist S., **Johnson J.W.**, Price-Whelan A., Sit T., Stone-Martinez A., Weinberg D.H.
2024, ApJ, 169, 280 – 297 arxiv:2410.22121
12. *Modeling the Galactic Chemical Evolution of Helium*
Weller M.K., Weinberg D.H., **Johnson J.W.**
2024, MNRAS, 583, 1517 – 1534 arxiv:2404.08765
13. *Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution*
Dubay L.O., Johnson J.A., **Johnson J.W.**
2024, ApJ, 973, 55 – 80 arxiv:2404.08059
14. *The APO-K2 Catalog. II. Accurate Stellar Ages for Red Giant Branch Stars Across the Milky Way*
Warfield J.T., et al., incl. **Johnson J.W.**
2024, AJ, 167, 208 – 231 arxiv:2403.03249
15. *Nature vs. Nurture: Distinguishing effects from stellar processing and chemical evolution on carbon and nitrogen in red giant stars*

- Roberts J.D., et al., incl. **Johnson J.W.**
2024, MNRAS, 530, 149 – 166 arxiv:2403.03249
16. *The Scale of Stellar Yields: Implications of the Measured Mean Iron Yield of Core Collapse Supernovae*
Weinberg D.H., Griffith E.J., **Johnson J.W.**, Thompson T.A.
2023, ApJ, 973, 122 – 136 arxiv:2309.05719
 17. *Untangling the Sources of Abundance Dispersion in Low-Metallicity Stars*
Griffith E.J., Johnson J.A., Weinberg D.H., Ilyin I., **Johnson J.W.**, Rodriguez-Martinez R., Strassmeier K.G.
2022, ApJ, 944, 47 – 67 arxiv:2210.01821
 18. *Birth of the Galactic Disk Revealed by the H3 Survey*
Conroy C., et al., incl. **Johnson J.W.**
2022, submitted to AAS Journals, under peer review arxiv:2204.02989
 19. *Primordial Helium-3 Redux: The Helium Isotope Ratio of the Orion Nebula*
Cooke R.J., Noterdaeme P., **Johnson J.W.**, Pettini M., Welsh L., Peroux C., Murphy M.T., Weinberg D.H.
2022, ApJ, 932, 60 – 76 arxiv:2203.11256
 20. *Residual Abundances in GALAH DR3: Implications for Nucleosynthesis and Identification of Unique Stellar Populations*
Griffith E.J., Weinberg D.H., Buder S., Johnson J.A., **Johnson J.W.**, Vincenzo F.
2021, ApJ, 931, 23 – 50 arxiv: 2110.06240
 21. *Chemical Cartography with APOGEE: Mapping Disk Populations with a Two-Process Model and Residual Abundances*
Weinberg D.H., et al., incl. **Johnson J.W.**
2021, ApJS, 260, 32 – 77 arxiv:2108.08860
 22. *CNO dredge-up in a sample of APOGEE/Kepler red giants: Tests of stellar models and galactic evolutionary trends of N/O and C/N*
Vincenzo F., et al., incl. **Johnson J.W.**
2021, submitted to MNRAS, under peer review arxiv:2106.03912
 23. *The Impact of Black Hole Formation on Population-averaged Supernova Yields*
Griffith E.J., Sukhbold T., Weinberg D.H., Johnson J.A., **Johnson J.W.**, Vincenzo F.
2021, ApJ, 921, 73 – 94 arxiv:2103.09837
 24. *Nucleosynthesis signatures of neutrino-driven winds from proto-neutron stars: a perspective from chemical evolution models*
Vincenzo F., Thompson T.A., Weinberg D.H., Griffith E.J., **Johnson J.W.**, Johnson J.A.
2021, MNRAS, 508, 3499 – 3507 arxiv:2102.04920
 25. *The Similarity of Abundance Ratio Trends and Nucleosynthetic Patterns in the Milky Way Disk and Bulge*
Griffith E.J., et al., incl. **Johnson J.W.**
2021, ApJ, 909, 77 – 101 arxiv:2009.05063